5.8.3 The Theory of Evolution and Natural Selection

After completing this chapter you should be able to relate the structure and function of living things to models, theories and laws.

For Questions 1 to 5 choose the best answer (A), (B), (C) or (D).

1. Consider this statement:

'Organisms who are able to survive well in an environment pass on their characteristics to more offspring and these characteristics become more common'.

What theory does this statement allude to?

(A) The theory of adaptations.
(B) The theory of genetic modification.
(C) The theory of natural selection.
(D) The theory of natural survival.

2. What causes variations in the genetic make-up of organisms from the same species?

(A) Mutations caused by external factors.
(B) Inherited characteristics from parents.
(C) Environmental factors.
(D) Both inherited and environmental factors.

3. What is the meaning of the term 'evolution'?

(A) To develop gradually.
(B) To adapt to one's surroundings.
(C) To change as mutations occur.
(D) To change as genes change.

4. Which of the following statements is *not* proposed by natural selection?

(A) Members within a single species show variation.
(B) Humans have slowly developed from apes.
(C) In the struggle to survive, some members of a species will die before they reproduce.
(D) Over time, organisms become better suited to their environment.

5. In what way does artificial selection provide evidence for natural selection?

(A) In deliberately isolating favoured characteristics in organisms and breeding new generations that contain those characteristics, it copies natural selection.
(B) It forces natural selection to occur over a much shorter period of time.
(C) It produces changes in the DNA of organisms by genetic engineering and this is similar to the way in which natural selection works.
(D) Artificial selection is a totally different process to natural selection and, as such, provides no evidence for natural selection.

6.

(a) Explain the concept of 'survival of the fittest'.

(b) What is another term for 'survival of the fittest'?

7.

(a) How does *artificial selection* differ from *natural selection*?

(b) A farmer had a flock of sheep that gave mixed quality wool. Some gave very high grade fleece, others quite poor fleece. Explain how the farmer could use artificial selection to improve the overall quality of the fleece produced by his flock.

(c) Convert your ideas from (b) into a flow chart to summarise the procedure you recommend.

8. Discuss one piece of evidence that scientists use to claim evolution is a correct theory.

9. Explain the importance of sexual reproduction in an evolutionary process.

10. Explain how the development of super bugs supports Darwin's idea of 'the survival of the fittest'.

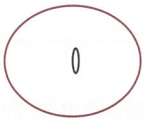
11. When Darwin examined the different life forms on the Galapagos Islands, one of his observations was that there was considerable variation in the size and shapes of the beaks of different species of finches.

Explain how the theory of evolution would account for this variation.

12. The diagrams show how natural selection works in a bacteria population.

A  Bacteria type X in culture medium.

B  Bacteria X multiply.
Some mutate to form new strain type Y.

C  Antibiotic added which kills all bacteria X.

D  Type Y bacteria multiply, unaffected by antibiotic.

(a) Explain how the information above is an example of natural selection.

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(b) When an organism reproduces, DNA mutations can occur. Explain how these can be helpful in the survival of the organism.

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(c) Recall the name of the thread-like structures that contain DNA.

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(d) In which part of a cell are these structures found?

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13. Describe the process of natural selection.

14. Outline the evidence for present-day organisms having developed from different organisms in the past.